# Cheng Chen

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Researcher in generative models of the physical world: models that simulate how scenes, objects and agents change under interaction, and that let robots and agents act on what they simulate. First-author work includes interactive video simulation (DiVA), 3D and CAD generation (CVPR 2024, 2025), world models that turn human video into robot experience (AnyWorld, CoRL 2026), and efficient VLA policies (SwiftVLA, CVPR 2026). Industry research at XPENG Robotics and Ant Group.

## EDUCATION

**Nanyang Technological University, Singapore** | Jan. 2023 - present
Ph.D. in Computer Science. Supervisors: Prof. Guosheng Lin and Dr. Fayao Liu.
Thesis submitted; defense expected December 2026; degree expected January 2027.

**Fudan University, Shanghai, China**
M.Eng. in Information Engineering, 2019 - 2022; B.Eng. in Electronic Engineering, 2015 - 2019.

## SELECTED RESEARCH CONTRIBUTIONS

### AnyWorld | First author | CoRL 2026

Turned unpaired human egocentric video into robot-native training experience by factorizing action, camera and embodiment control in a video world model. Used the model to generate targeted counterfactual experience that repaired real-robot policy failures: it removed a systematic false-completion behavior and enabled instruction-conditioned target selection under identical observations. Project / paper.

### Physical Probing for Frontier-Model Robot Agents | First author | In preparation, 2026

Frontier agents such as GPT-6 can plan robot tasks but lack physical properties that are not observable from pixels, such as mass, friction and compliance. Proposed a learned, task-shared probing policy that decides which physical interactions to perform, and when to stop, to gather decision-sufficient evidence for a frozen frontier agent at minimal physical cost. Introduced the Physical Adaptation Cost (PAC) benchmark.

### SwiftVLA | Co-first author | CVPR 2026

Lightweight vision-language-action model that injects spatiotemporal geometry representations, connecting visual geometry and temporal dynamics to efficient robot-action generation at minimal overhead. Paper.

### XPACE | Co-author | XPENG Robotics technical report, 2026

Company-scale unified model for joint future-video and executable-action prediction and action-conditioned simulation, trained with joint flow matching, chunk-wise autoregressive generation and self-gradient forcing. Built world-model training and the heterogeneous robot and human data pipeline, including full-parameter fine-tuning of 10B+ video models. Project / paper.

### DiVA | First author | Under review at IJCV

Interactive video simulation across multi-turn interactions. Developed Anchored Video Continuation to connect action, waiting and transition segments while preserving identity and spatial consistency, and characterized the trade-off between visual quality and interaction latency. Paper.

## RESEARCH EXPERIENCE

**XPENG Robotics** | Research Intern | Dec. 2025 - present
World-model training and data for XPACE; first-author AnyWorld, from experience generation to real-robot policy interventions.

**Ant Group** | Research Intern | Apr. 2025 - Oct. 2025
Interactive video generation and editing; first-author DiVA.

**Nanyang Technological University** | Ph.D. researcher | Jan. 2023 - present
Generative 3D and CAD modeling (Sculpt3D, CADCrafter), then embodied agents: physical probing for frontier-model robot agents, and slow-fast execution for LLM agents acting in real time.

**Research engineering:** Full-parameter fine-tuning of 10B+ video models; multi-node distributed training.

**Available:** February 2027.

## SELECTED PUBLICATIONS AND PREPRINTS

**AnyWorld: Factorized Egocentric World Models for Cross-Embodiment Generalization | First author**
**Cheng Chen** et al. *CoRL 2026.*

**SwiftVLA: Unlocking Spatiotemporal Dynamics for Lightweight VLA Models at Minimal Overhead | Co-first author**
Chaojun Ni, **Cheng Chen** (co-first author), et al. *CVPR 2026.*

**HoloCine: Holistic Generation of Cinematic Multi-Shot Long Video Narratives | Co-author**
Yihao Meng, Hao Ouyang, Yue Yu, Qiuyu Wang, Wen Wang, Ka Leong Cheng, Hanlin Wang, Yixuan Li, **Cheng Chen**, et al. *CVPR 2026, Highlight.*

**iMontage: Unified, Versatile, Highly Dynamic Many-to-many Image Generation | Co-author**
Zhoujie Fu, Xianfang Zeng, Jinghong Lan, Xinyao Liao, **Cheng Chen**, et al. *CVPR 2026.*

**The World is Your Canvas: Painting Promptable Events with Reference Images, Trajectories, and Text | Co-author**
Hanlin Wang, Hao Ouyang, Qiuyu Wang, Yue Yu, Yihao Meng, Wen Wang, Ka Leong Cheng, Shuailei Ma, Qingyan Bai, Yixuan Li, **Cheng Chen**, et al. *SIGGRAPH Asia 2026.*

**XPACE: Joint World and Action Modeling from Heterogeneous Experience | Co-author**
Jiacheng Wei, Jerry Bai, Xiaoyu Yue, Zidong Wang, Xiaoyang Guo, **Cheng Chen**, et al. *arXiv preprint, 2026.*

**DiVA: Enabling Interactive Digital Life Simulation via Video Models | First author**
**Cheng Chen** et al. *arXiv preprint, 2026; under review at IJCV.*

**CADCrafter: Generating Computer-Aided Design Models from Unconstrained Images | First author**
**Cheng Chen** et al. *CVPR 2025.*

**ADHMR: Aligning Diffusion-based Human Mesh Recovery via Direct Preference Optimization | Co-author**
Wenhao Shen, Wanqi Yin, Xiaofeng Yang, **Cheng Chen**, et al. *ICML 2025.*

**Text-to-Image Rectified Flow as Plug-and-Play Priors | Co-author**
Xiaofeng Yang, **Cheng Chen**, Xulei Yang, Fayao Liu, Guosheng Lin. *ICLR 2025.*

**Sculpt3D: Multi-View Consistent Text-to-3D Generation with Sparse 3D Prior | First author**
**Cheng Chen** et al. *CVPR 2024.*

**Learn to Optimize Denoising Scores: A Unified and Improved Diffusion Prior for 3D Generation | Co-author**
Xiaofeng Yang, Yiwen Chen, **Cheng Chen**, et al. *ECCV 2024.*

**UTC: A Unified Transformer with Inter-Task Contrastive Learning for Visual Dialog | First author**
**Cheng Chen** et al. *CVPR 2022.*

## TALKS, SERVICE AND MENTORING

**Invited talks:** AnyWorld and XPACE, 具身智能之心, 2026; AnyWorld, AnySyn3D, 2026.

**Top Reviewer**, NeurIPS 2026. **Conference reviewer:** NeurIPS, ICLR, CVPR, ICML, ICCV, ECCV, ACM MM, CoRL.

**Mentoring:** supervising a graduate student on a video-generation paper targeting ICML 2027.

## AWARDS AND HONOURS

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| **SINGA Scholarship, Singapore International Graduate Award** | Jan. 2023 |
| **Outstanding Graduate of Fudan University** | Apr. 2022 |
| **National Scholarship (1/72)** | Oct. 2021 |
| **First-Class Scholarship, Fudan University** | Oct. 2019 |
| **Outstanding Student, Information School of Fudan University** | May 2017 |

## SKILLS

**Research:** Video world models, interactive video simulation, 3D/CAD generation; human-to-robot transfer, VLA policies, real-robot evaluation; frontier-model embodied agents.

**Training:** Large video-model fine-tuning, multi-node distributed training, heterogeneous robot and human data pipelines, autoregressive generated-context learning.

**Tools:** Python, PyTorch, TensorFlow, Linux, LaTeX. **Languages:** Mandarin Chinese, English.